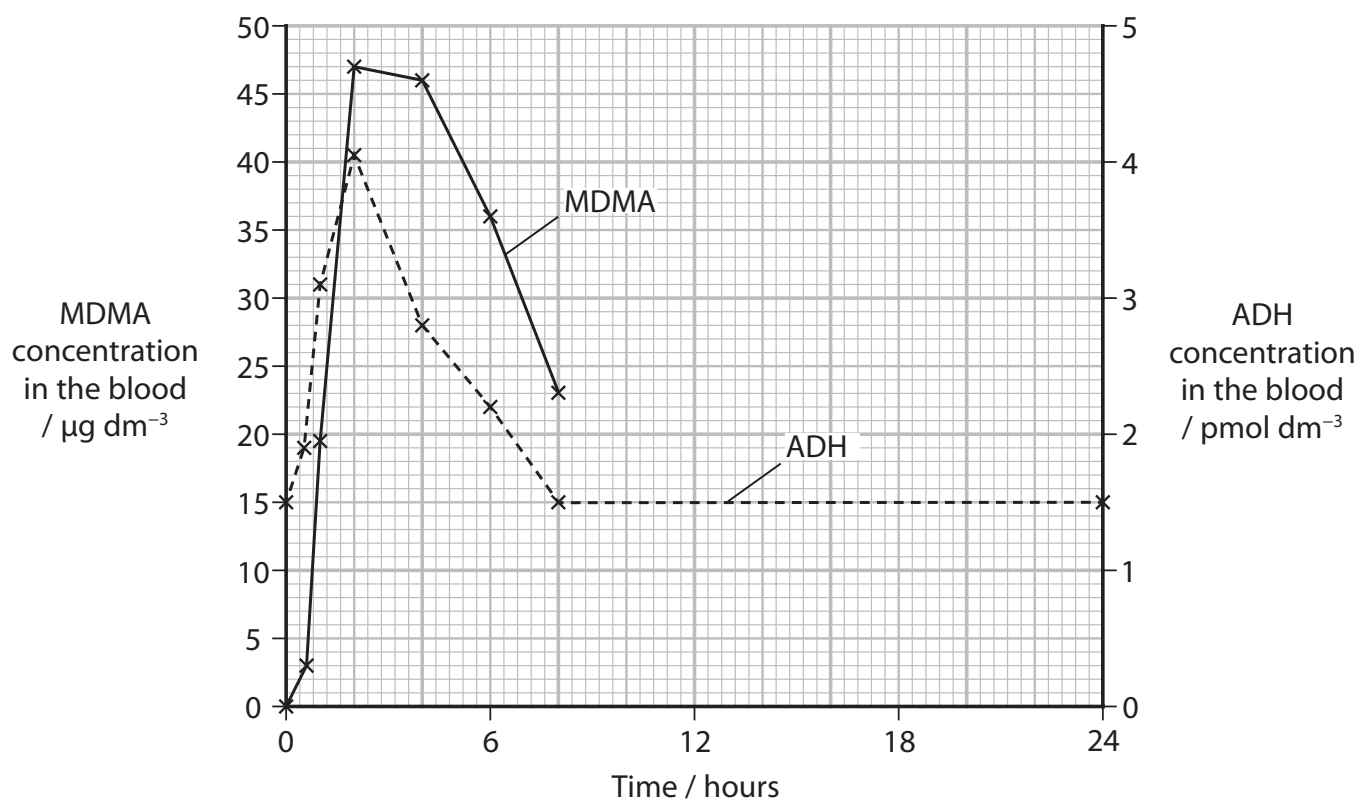


3 Ecstasy (MDMA) is a recreational drug that can have serious toxic effects.

(a) The effect of MDMA on ADH concentration in blood has been studied.

In one study, each person in a group of volunteers was given MDMA. The MDMA and ADH concentrations in the blood were measured.

The graph shows the results of this study.



(i) Estimate how long it takes the MDMA concentration in the blood to become $0.0 \mu\text{g dm}^{-3}$.

(1)

Answer hours

(ii) Calculate the rate of removal of ADH from the blood between 4 and 8 hours.

(2)

Answer

*(b) Taking MDMA causes a person to be very thirsty. This can lead to swelling of the brain, which can be fatal.

The table shows some laboratory results for one case in which a person suffered swelling of the brain.

Time since taking MDMA / hours	ADH concentration / pmol dm^{-3}	Sodium ion concentration in the blood / mmol dm^{-3}	Appearance of the brain in computed tomography (CT)
9	4.5	112	Swollen
96	1.2	131	Normal

Comment on the role of ADH in MDMA-induced brain swelling.

Use the information in the graph and table as well as your own knowledge to support your answer.

(6)

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(Total for Question 3 = 9 marks)